Title: Technical Brief: JARVIS v7.0 (Agentic Voice Assistant)

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1. Objective: To develop a high-throughput, fully autonomous agentic voice assistant featuring ultra-low latency cloud-hosted inference and non-blocking asynchronous local tool execution loops.

2. Architecture:

- **Core Brain:** Llama 3.3 70B state-of-the-art model hosted via Groq Cloud API for ultra-fast streaming and native function routing.
- **Voice Engine:** Multi-threaded asynchronous audio processing loop (STT/TTS pipeline) running on Python's `asyncio` to eliminate interface freezing.
- **Routing Loop:** Native tool-calling framework for autonomous local system execution and live API management (Navigation, Stocks, Email drafts).
- **UI Framework:** High-performance production-ready Terminal User Interface (TUI) via Rich to completely minimize system resource overhead.

3. Key Performance:

- **Latency:** Sub-50ms core event loops with lightning-fast token generation speeds.
- **Execution Safety:** Deterministic JSON schema verification with automatic recursion to capture and heal tool execution errors instantly.

4. Empirical Validation:

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  IIT Roorkee Young Coders Hackathon: Secured **2nd Runner-up** position out of 500+ competing teams. Evaluated rigorously on runtime script optimization, latency control, and multi-threaded system stability.

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5. Conclusion: JARVIS v7.0 demonstrates that massive, cloud-scale LLMs (70B) can be seamlessly integrated with localized asynchronous structures to deploy production-grade autonomous agents without local hardware constraints.